

Fred Zirbel

Principal Security Analyst | Detection Engineering | Incident Response

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Summary

Principal Security Analyst leading high-velocity investigations across identity, email, endpoint, and cloud telemetry. Sustains a 16-minute average time to investigation across 300+ monthly investigations; engineered 500+ suppression filters to reduce recurring noise. Builds open-source investigation and detection tooling in Python.

Experience

Principal Security Analyst

Jul 2026 - Present

Critical Start, Plano, TX

- Own phishing, malware, and identity investigations and coordinated remediation in MDR operations supporting 2,500+ customer environments and 30+ integrated security products.
- Sustain a 16-minute average TTI across 300+ monthly investigations; guide customer response decisions and mentor analysts on evidence quality, scope, and escalation readiness.

Senior Security Analyst

Jul 2025 - Jul 2026

Critical Start, Plano, TX

- Correlated email, identity, network, and endpoint telemetry with custom KQL to reconstruct attacker actions and scope cross-tenant campaigns.
- Engineered and quality-reviewed 500+ KVP and regex suppression filters, eliminating thousands of recurring false positives.

Security Analyst

Oct 2024 - Jul 2025

Critical Start, Plano, TX

- Triaged identity, phishing, malware, and endpoint alerts across Defender, Sentinel, CrowdStrike, and SentinelOne; produced client-ready escalations and remediation guidance.

Selected Projects

SOC Box | Python, FastAPI, Playwright, Docker, KQL

GitHub

Self-hosted analyst workspace for eight-dimension URL detonation, streamed results, IP enrichment, bulk analysis, and Defender XDR/Sentinel hunting-query generation. Includes OIDC/service-token auth, SSRF protection, rate limiting, and automated security checks.

SIGIL | Python, Sigma, Splunk, Sentinel, MITRE ATT&CK

GitHub

Detection-as-code pipeline that validates Sigma schemas, converts to SPL/KQL, tests rules against sample logs, and maps coverage gaps across the Enterprise ATT&CK matrix.

Skills

Investigation: Microsoft Defender XDR, Sentinel, CrowdStrike Falcon, Cortex XDR, SentinelOne, Splunk ES, Sumo Logic

Detection and engineering: KQL, Sigma, MITRE ATT&CK, Python, FastAPI, Docker, GitHub Actions, OpenSearch

Education and Certifications

M.S., Cybersecurity and Information Assurance, Western Governors University

B.S., Information Systems, The

University of Texas at Arlington

CompTIA SecurityX (CASP+), CySA+, PenTest+, Security+; ISC2 Certified in Cybersecurity

CISM - In Progress